# Simplified Expression for Ziv–Zakai Bound in MIMO Radar Systems

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***Abstract*—Although the Ziv-Zakai Bound (ZZB) can predict the most favorable performance of MSE in the range of all signal-to-noise ratios (SNR), its computational complexity is often unbearable. The main reason is that a complex optimization problem will be encountered when evaluating ZZB of the multi-parameter joint estimation. This paper presents the result of the optimization problem in numerical form, thereby further deriving the simplified expression of ZZB in the context of joint parameter estimation in MIMO radar systems. This expression can significantly reduce the computational complexity of ZZB, so that preliminary system design can be carried out efficiently.**

***keywords*—Simplified expression, MIMO radar, Ziv-Zakai bound (ZZB)***

## I. Introduction

The performance of the estimator is generally measured by the corresponding mean squared error (MSE) of parameter estimation. When the observation is affected by noise and it is difficult to obtain the exact mean square error, the lower bounds of the MSE are useful indicators. These lower bounds play an important role in evaluating the performance of different estimators. They can not only serve as a baseline for analyzing the limitations of parameter estimation problems, but also be used as a function of the system parameters to measure the performance of the estimation [1].

Recently, various lower bounds are widely used for the problems of signal parameter estimation in radar systems where the unknown parameters are embedded in the signal received in a noisy environment. The most common unknown parameters are the frequency, time-delay or position and speed of a target. In this type of estimation problems, it can usually be divided into three obvious regions, named the asymptotic region with long observation time and high signal-to-noise ratio, the nonasymptotic region (also called a priori performance region) with small observation time and low signal-to-noise ratio, and an ambiguity region between them [2]. Local bounds, such as Cramer–Rao bound (CRB) [3] [4], which is a commonly used bound belonging to the deterministic covariance inequality bounds family [5]. It is easier to evaluate than other general bounds. CRB provides more accurate predictions for MSE performance in the asymptotic region. However, for the problem of nonlinear estimation, its predictions of MSE in the other two regions are not accurate. The CRB starts to show deviation in the ambiguity region, and in the nonasymptotic region its results have almost no reference value. In addition, CRB only determines the lower bound of the unbiased estimator, and does not use the prior information of the estimated parameter.

There is also Barankin bound [6] that is in the same family as CRB, which treats the parameter to be estimated as a deterministic unknown quantity. Although it is tighter than CRB and can get more reasonable MSE prediction results in the ambiguity area, it needs to be optimized on several test points, which contribute to a high computational complexity.

The Ziv-Zakai bound (ZZB) is a more powerful global bound that performs well in the entire range of signal-to-noise ratio. The literature illustrates the research of ZZB in various estimation problems [7]–[11]. As a Bayesian bound, it can use the prior distribution of the parameter to be estimated. ZZB is suitable for a variety of scenarios including unbiased estimation, and there are no strict restrictions on estimation parameters. In a nutshell, ZZB combines the parameter estimation problem with the binary hypothesis testing problem while connecting the MSE in the estimation problem with the average error probability in the binary hypothesis testing problem. Compared with CRB, ZZB provides similar predictions in asymptotic region, and starts to show different results from CRB in ambiguity region, and finally has far better results in nonasymptotic region.

In order to conduct preliminary system design, it is necessary to efficiently compare the performance of different systems. However, the evaluation of ZZB is very time-consuming. The main reason is that this evaluation in the joint parameter estimation problem involves an optimization problem of multiple integrals. This paper starts with the joint location and velocity estimation problem in non-coherent MIMO radar systems. Our simulation of ZZB focuses on the analysis of the optimization problem about its multiple integrals. Finally, a simplified expression for ZZB is derived and this conclusion is verified in different scenarios.

## II. Brief Introduction of ZZB

First, we briefly review the formula and derivation of ZZB. The estimated value of the parameter $\boldsymbol{\theta}$ is expressed as $\hat{\boldsymbol{\theta}}$ the matrix of mean estimation error (MSE) is defined

$$\boldsymbol{\Psi} = E\{\left|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\right|^2\} \tag{1}$$

By deriving the lower bound of the quadric form of MSE, using the properties of mean value and other deformations of the integral variables [2], the calculation of MSE can be finally transformed into the calculation of the following formula

$$P_e(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma}) = Pr(\boldsymbol{v}^\dagger\hat{\boldsymbol{\theta}} > \frac{h}{2} + \boldsymbol{v}^\dagger\boldsymbol{\varphi_0} | \boldsymbol{\theta} = \boldsymbol{\varphi_0})P_0 + Pr(\boldsymbol{v}^\dagger\hat{\boldsymbol{\theta}} \leq \frac{h}{2} + \boldsymbol{v}^\dagger\boldsymbol{\varphi_0} | \boldsymbol{\theta} = \boldsymbol{\varphi_0}+\boldsymbol{\sigma})P_1 \quad (2)$$

where $P_0 = p_{\boldsymbol{\theta}}(\boldsymbol{\varphi_0})/[p_{\boldsymbol{\theta}}(\boldsymbol{\varphi_0}) + p_{\boldsymbol{\theta}}(\boldsymbol{\varphi_0}+\boldsymbol{\sigma})]$ and $P_1 = 1 - P_0$, $\boldsymbol{v}^\dagger$ means the transpose of $\boldsymbol{v}$, and $\boldsymbol{v}^\dagger\boldsymbol{\sigma} = h$.

Now we can associate the estimated error $\left|\boldsymbol{v}^\dagger(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})\right|$ with the average error probability $P_e(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$ of a binary hypothesis test problem. This binary hypothesis problem is defined as follows

$$\begin{aligned} &H_0 : \boldsymbol{\theta} = \boldsymbol{\varphi_0}; \boldsymbol{r} \sim p_{\boldsymbol{r}|\boldsymbol{\theta}}(\boldsymbol{r}|\boldsymbol{\varphi_0}) \\ &H_1 : \boldsymbol{\theta} = \boldsymbol{\varphi_0}+\boldsymbol{\sigma}; \boldsymbol{r} \sim p_{\boldsymbol{r}|\boldsymbol{\theta}}(\boldsymbol{r}|\boldsymbol{\varphi_0}+\boldsymbol{\sigma}) \end{aligned} \quad (3)$$

where $\boldsymbol{r}$ are the noisy observations, with prior probability $Pr(H_0) = P_0, Pr(H_1) = P_1$. Adopt the suboptimal decision rule

$$\begin{aligned} &\text{Decide } H_0 \text{ when } \boldsymbol{v}^\dagger\hat{\boldsymbol{\theta}} \leq \tfrac{h}{2} + \boldsymbol{v}^\dagger\boldsymbol{\varphi_0} \\ &\text{Decide } H_1 \text{ when } \boldsymbol{v}^\dagger\hat{\boldsymbol{\theta}} > \tfrac{h}{2} + \boldsymbol{v}^\dagger\boldsymbol{\varphi_0} \end{aligned} \quad (4)$$

Then $P_e(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$ is the average error probability of the suboptimal decision problem. The minimum error probability can be obtained through the likelihood ratio test, that is, the minimum value of $P_e(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$ is $P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$. Then by adding valley-fifilling function [12] and some other manipulation, these are also discussed concretely in [13], we can get the final ZZB

$$\boldsymbol{v}^\dagger\boldsymbol{\Psi}\boldsymbol{v} \geq \frac{1}{2}\int_0^\infty \nu\{\max_{\boldsymbol{\sigma}:\boldsymbol{v}^\dagger\boldsymbol{\sigma}=h}\int_{\boldsymbol{\Theta}}[p_{\boldsymbol{\theta}}(\boldsymbol{\varphi_0}) + p_{\boldsymbol{\theta}}(\boldsymbol{\varphi_0}+\boldsymbol{\sigma})] \times P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})d\boldsymbol{\varphi_0}\}hdh \quad (5)$$

## III. Simplified Expression in MIMO Radar Systems

ZZB provides an accurate error bound under all regions, but it requires complex calculations. In this article we mainly focus on the estimation problem of MIMO radar where the computational complexity of ZZB is more notable, and try to reduce this complexity.

Considering a MIMO radar system [13], it has *M* transmitters and *N* receivers on the $(x, y)$ plane $(x_{tm}, y_{tm})$, *m*=1,. . . , *M* and $(x_{rn}, y_{rn})$, *n*=1,. . . ,*N* respectively, both of which are distributed in a sector with the origin of a Cartesian coordinate system. Supposing a target is at an unknown position $\boldsymbol{p} = (x, y)^\dagger$ locating in the $x - y$ plane with reference to the system and moves with an unknown speed represented by its projections on the $x$ and $y$ axes $\boldsymbol{v} = (v_x, v_y)^\dagger$. The *m*-th transmitter sends a narrow-band detection pulse with duration *T*. The complex envelope of this pulse is $\sqrt{\varepsilon/M}s_m(t)$, where $s_m(t)$ is the low-pass equivalent expression of the transmitted signal and $\varepsilon$ is the total transmission energy. $\boldsymbol{\varphi_0}$ is $(x, y, v_x, v_y)^\dagger$, and $\boldsymbol{\sigma}$ is $(\sigma_x, \sigma_y, \sigma_{v_x}, \sigma_{v_y})^\dagger$. Then the complex envelope of signal sent by the *m*-th transmitter and received by the *n*-th receiver is

$$r_{mn}(t) = \sqrt{\frac{\varepsilon}{M}}a_{mn}s_{mn}(t - \bar{\tau}_{mn}(\boldsymbol{p}))e^{j2\pi\bar{f}_{mn}(\boldsymbol{p},\boldsymbol{v})t} + w_{mn}(t) \quad (6)$$

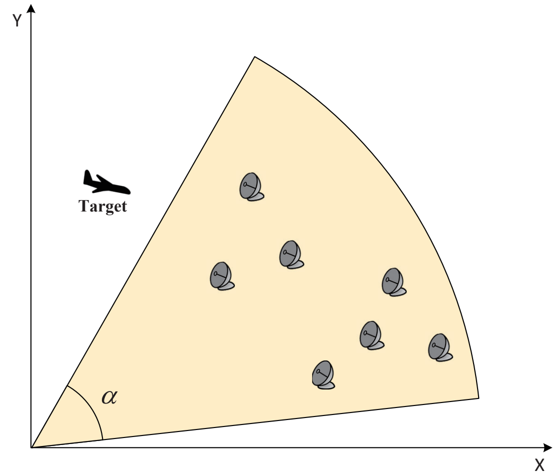



Fig. 1. Configuration of the MIMO radar system.

where $a_{mn}$ and $w_{mn}(t)$ represent the target reflection coefficient and additive clutter-plus-noise of the *mn*-th signal transmission path respectively. $\bar{f}_{nm}(\boldsymbol{p}, \boldsymbol{v})$ and $\bar{\tau}_{nm}(\boldsymbol{p})$ denote the doppler shift and time delay with respect to the mn-th path and a target moving with velocity $\boldsymbol{v}$ at the position $\boldsymbol{p}$.

Now make assumptions about the above parameters to get a simplified but useful example. Assume that the reflection coefficient $a_{mn}$ of the target is complex Gaussian random variable with zero mean and independent of each other. For all *m* and *n*, $a_{mn}$ has the same variance $\sigma_{a_{mn}} = \sigma_a$. Additive clutter-plus-noise $w_{nm}(t)$ obeys zero mean, complex Gaussian distribution with known variance $\sigma_{w_{mn}}$. For any different signal transmission path and different time, $w_{nm}(t)$ is mutual independent. Besides, the variance of it of each path is the same, that is, $\sigma_{w_{mn}} = \sigma_w$.

In addition, assume that the transmission signal $s_m(t)$, *m*=1,. . . , *M* has unit energy, and for any different transmitter *m*, the time delay $\tau$, and doppler shift $f$, $s_m(t)$ maintain orthogonality. $x, y, v_x, v_y$ are assumed to be independent, where $x, y$ are uniformly distributed in $[0, D]$, $v_x, v_y$ are uniformly distributed in $[0, V]$. For the hypothesis of equally likely possibility, there is $Pr(H_0) = Pr(H_1) = \frac{1}{2}$.

Under these assumptions, the expression of ZZB for the target coordinates $(x, y)$ and speed $(v_x, v_y)$ can be expressed as

$$\begin{aligned} &E\{|x - \hat{x}|^2\} = \boldsymbol{v}_x^\dagger\boldsymbol{\Psi}\boldsymbol{v_x} = \{\boldsymbol{\Psi}\}_{11} \\ &\geq \frac{1}{D^2V^2}\int_0^D (\max_{\delta_y,\delta_{vx},\delta_{vy}}\int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})d\boldsymbol{\varphi_0})\delta_x d\delta_x \\ &\triangleq ZZB_x \end{aligned} \quad (7)$$

$$\begin{aligned} &E\{|y - \hat{y}|^2\} = \boldsymbol{v}_y^\dagger\boldsymbol{\Psi}\boldsymbol{v_y} = \{\boldsymbol{\Psi}\}_{22} \\ &\geq \frac{1}{D^2V^2}\int_0^D (\max_{\delta_x,\delta_{vx},\delta_{vy}}\int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})d\boldsymbol{\varphi_0})\delta_y d\delta_y \\ &\triangleq ZZB_y \end{aligned} \quad (8)$$

$$E\{|v_x - \hat{v}_x|^2\} = \boldsymbol{v}_{vx}^{\dagger}\boldsymbol{\Psi}\boldsymbol{v_{vx}} = \{\boldsymbol{\Psi}\}_{33}$$
$$\geq \frac{1}{D^2V^2}\int_0^V (\max_{\delta_x,\delta_y,\delta_{vy}} \int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0} + \boldsymbol{\sigma})d\boldsymbol{\varphi_0})\delta_{vx}d\delta_{vx}$$
$$\triangleq ZZB_{vx} \tag{9}$$

$$E\{|v_y - \hat{v}_y|^2\} = \boldsymbol{v}_{vy}^{\dagger}\boldsymbol{\Psi}\boldsymbol{v_{vy}} = \{\boldsymbol{\Psi}\}_{44}$$
$$\geq \frac{1}{D^2V^2}\int_0^V (\max_{\delta_x,\delta_y,\delta_{vx}} \int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0} + \boldsymbol{\sigma})d\boldsymbol{\varphi_0})\delta_{vy}d\delta_{vy}$$
$$\triangleq ZZB_{vy} \tag{10}$$

where

$$\boldsymbol{\Theta} = \{x, y, v_x, v_y\,|x \in [0, D-\delta_x], y \in [0, D-\delta_y]$$
$$v_x \in [0, V-\delta_{vx}], v_y \in [0, V-\delta_{vy}]\}$$

As a MIMO radar joint parameter estimation problem, many parameter settings in ZZB evaluation can reduce the complexity and still get useful results. For example, the parameter $\boldsymbol{\theta}$ is considered to be uniformly distributed, and the reflection coefficient $a_{mn}$ is assumed to be simpler Gaussian distribution, etc. But no matter how to simplify them, there is always an obstacle we have to deal with: the optimization problem of multiple integration $\max\limits_{\boldsymbol{\sigma}:\boldsymbol{v}^{\dagger}\boldsymbol{\sigma}=h}\int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})d\boldsymbol{\varphi_0}$. Considering that the expression of $P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$ is particularly complicated, this optimization problem is actually the main factor leading to the time-consuming evaluation of ZZB, so we try to simplify the evaluation of ZZB starting with this problem.

Take $ZZB_x$ as an example, and calculate (7), in which we use the Monte Carlo integration method to calculate the inner multiple integrals, and use simulated annealing [14], genetic algorithm [15],and other optimization algorithms to calculate the optimization problem $\max\limits_{\boldsymbol{\sigma}:\boldsymbol{v}^{\dagger}\boldsymbol{\sigma}=h}\int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})d\boldsymbol{\varphi_0}$. In the simulation process, we observe the changes about the values of the three variables $\sigma_y, \sigma_{v_x}, \sigma_{v_y}$ during the iteration of the simulated annealing algorithm, and count the final results of several optimization algorithms. Simulation results are shown below.

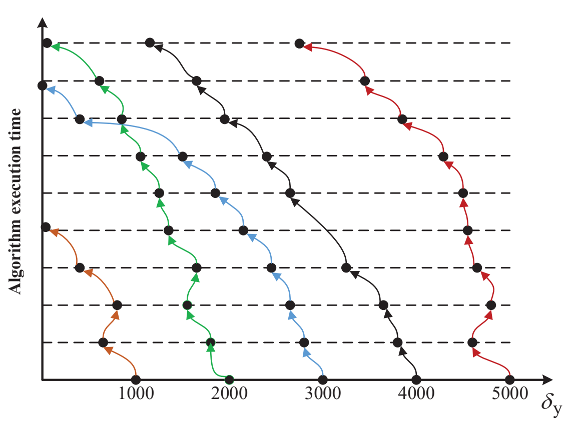



Fig. 2. Changes in $\sigma_y$ during simulated annealing.

Fig. 2 shows the evolution of $\sigma_y$ in the simulated annealing algorithm. $\sigma_{v_x}, \sigma_{v_y}$ and $\sigma_y$ have similar performance. This

TABLE I. Statistics of Simulated Annealing Algorithm Results

| | $\delta_y$ | $\delta_{vx}$ | $\delta_{vy}$ |
|---|---|---|---|
| $\delta_i$ | 238 | 236 | 238 |
| $0<\delta_i \leq 100$ | 2 | 5 | 4 |
| $100<\delta_i \leq \max(\delta_i) \leq 100$ | 2 | 1 | 1 |

TABLE II. Statistics of Particle Swarm Algorithm Results

| | $\delta_y$ | $\delta_{vx}$ | $\delta_{vy}$ |
|---|---|---|---|
| $\delta_i$ | 236 | 232 | 236 |
| $0<\delta_i \leq 100$ | 4 | 9 | 4 |
| $100<\delta_i \leq \max(\delta_i) \leq 100$ | 2 | 1 | 2 |

TABLE III. Statistics of Genetic Algorithm Results

| | $\delta_y$ | $\delta_{vx}$ | $\delta_{vy}$ |
|---|---|---|---|
| $\delta_i$ | 233 | 236 | 234 |
| $0<\delta_i \leq 100$ | 5 | 5 | 4 |
| $100<\delta_i \leq \max(\delta_i) \leq 100$ | 2 | 3 | 3 |

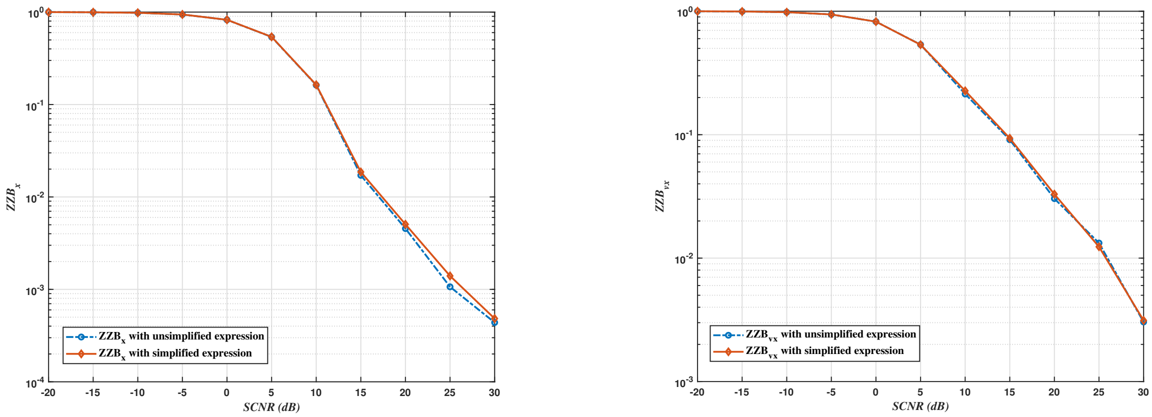
figure illustrates the specific operation process of the optimization algorithm. The transverse axis represents the selected initial value of $\sigma_y$. Here, five different initial values are used for five annealing processes. For each point in the figure, the abscissa is the value of $\sigma_y$ during the optimization process, and the ordinate is the running time of the optimization algorithm (not all shown here). It can be seen that as the optimization progresses, the number of algorithm's iterations increases, and $\sigma_y$ with different initial values gradually tend to 0. This result provides a good clue for the simplification of ZZB evaluation.

TABLE I, II, III count the final values of the three variables $\sigma_y, \sigma_{v_x}, \sigma_{v_y}$ obtained by different optimization algorithms when calculating $ZZB_x$. The evaluation of $ZZB_x$ needs to optimize the inner integral at different values of $\sigma_x$, so one calculation will face a lot of optimization problems. It can be clearly seen from the table that for all optimization algorithms, the three variables $\sigma_y, \sigma_{v_x}, \sigma_{v_y}$ almost all achieve zero values. For those few cases where the final value is not taken at 0, it is analyzed that this is caused by the randomness of the optimization algorithm itself and the lack of strict enough termination conditions. Similar results are obtained when calculating $ZZB_y, ZZB_{v_x}, ZZB_{v_y}$. So far we have reached a conclusion: in the MIMO radar joint parameter estimation problem, the solution of optimization problem in ZZB evaluation is obtained when the three independent variables all take the value 0.

*Definition 1:* Based on this, we can get the simplified expression of ZZB, $ZZB_x$ for instance

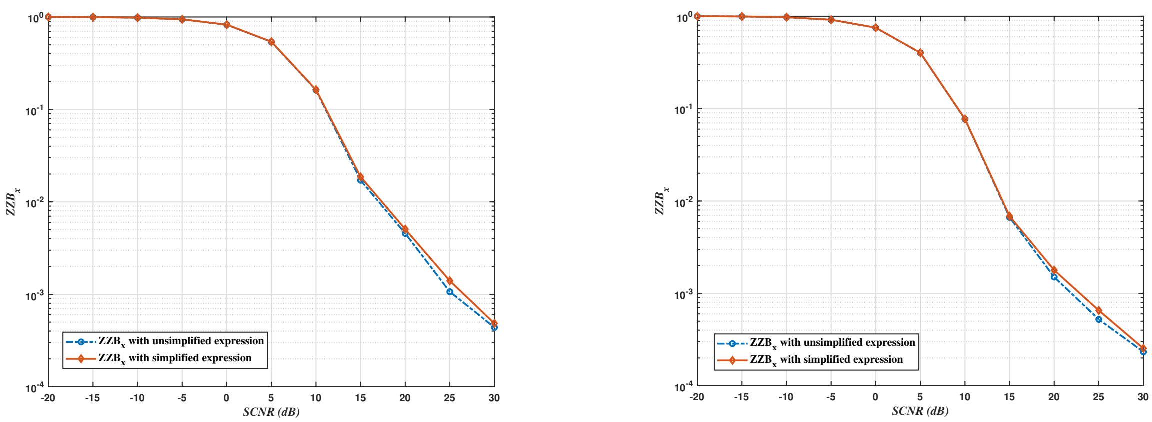
$$E\{|x - \hat{x}|^2\} = \boldsymbol{v}_x^{\dagger}\boldsymbol{\Psi}\boldsymbol{v_x} = \{\boldsymbol{\Psi}\}_{11}$$
$$\geq \frac{1}{D^2V^2}\int_0^D \int_{\boldsymbol{\Theta}} P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0} + \sigma_x)d\boldsymbol{\varphi_0}\sigma_x d\sigma_x \tag{11}$$

Since the expression $P_{\min}(\boldsymbol{\varphi_0}, \boldsymbol{\varphi_0}+\boldsymbol{\sigma})$ is too complicated, it is difficult to theoretically derive an analytical solution for this optimization problem. The concrete proof of this



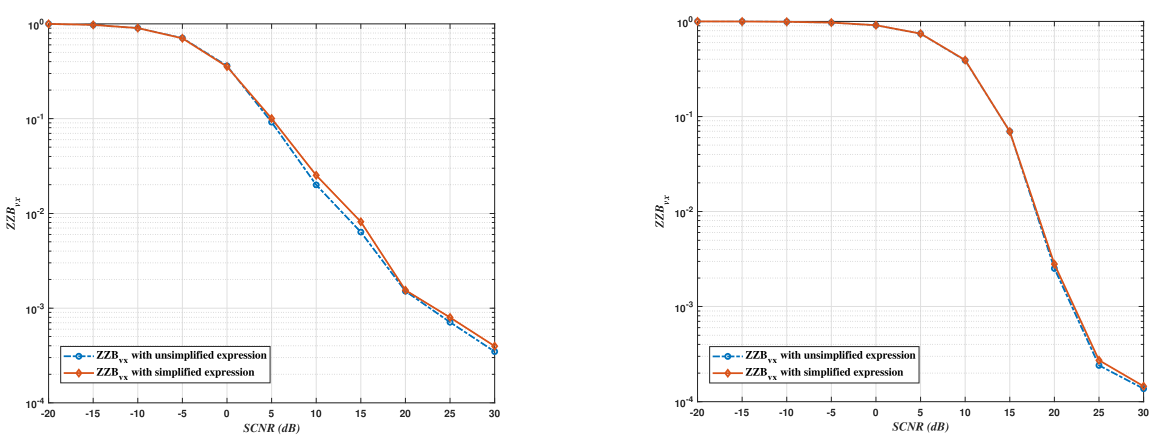

(a). $ZZB_x$ (b). $ZZB_{v_x}$

Fig. 3. $ZZB_x$, $ZZB_{v_x}$ (normalized) obtained by two expressions.




(a). 2 transmitting antennas and 3 receiving antennas (b). 1 transmitting antennas and 6 receiving antennas

Fig. 4. $ZZB_x$ (normalized) obtained by two expressions under different MIMO radar configurations.




(a). Gaussian pulses (b). Linear frequency modulated (LFM) pulses

Fig. 5. $ZZB_x$ (normalized) obtained by two expressions under different transmission signals.

conclusion is still in progress. Next, we verify the correctness and generality of the above conclusions in multiple scenarios.

## IV. Verification in Multiple Scenarios

In this section, the simplified expression obtained above is verified in different scenarios. Assume that $D$=5km, $V$=200m/s, transmitters and receivers are placed in a sector of $^{\pi}/_{3}$ with the origin of the coordinates as the vertex angle. Due to space limitations, other parameters will not be described in detail, the specific settings are similar to [13]. In the following various scenarios, we use simplified expressions and unsimplified expressions to evaluate ZZB separately, then compare the results obtained by them to verify our conclusion.

Fig. 3 shows the ZZB vs. signal-to-clutter-plus noise ratio (SCNR) curve of the target coordinate $x$ and the velocity $v_x$. The transmission signal used here is a set of coded orthogonal frequency division multiplexing (COFDM) signals [16], where the pulse width is set to $T = 2.5\mu s$, each waveform is composed of 16 sub-carriers with each sub-carrier having 16 binary symbols.

Where the red solid line is the result obtained by the simplified expression, and the blue dashed line is obtained by the unsimplified expression. It can be clearly observed that the two curves are almost overlapped. At higher signal-to-noise ratio, the results calculated by the two methods have small gaps. This is because Monte Carlo integration is used in the simulation process, so the results have certain volatility under high signal-to-noise ratio due to the limit on the number of stochastic points. Besides, the iterative process of the optimization algorithm itself has random characteristics, the results of which are also related to the strict degree of the algorithm's termination condition, so there is sometimes an error between the true global optimal solutions and the results. When we increase the number of Monte Carlo random cast points and impose stricter restrictions on the optimization algorithm, the gap between the two curves reduce, but this will greatly increase the evaluation complexity and eventually reach an unbearable level. Due to time constraints, we do not continue to perform more exact simulation calculations to get more accurate results.

Fig. 4 shows the results obtained when the number of transmitting antennas and receiving antennas are 2, 3 (Fig.4(a)) and 1, 6 (Fig.4(b)), respectively, under COFDM signal. It can be seen that the results obtained by the simplified expression and the unsimplified expression are also very consistent when the setting of MIMO radar is changed. The reason for the small deviation under the high signal-to-noise ratio has been explained in the description part of Fig. 3.

Fig. 5 illustrates the calculation results of $ZZB_x$ when the transmitted signals are single Gaussian pulses (Fig.5(a)) and linear frequency modulated (LFM) pulses (Fig.5(b)), where the time width of Gaussian signal is 0.1s,and the pulse of duration of LFM signal is $5\mu s$ as well as the time bandwidth product is 5. It can be seen from the results that our conclusions also perform well under different transmission signals, the results obtained by the two expressions are properly consistent. Similar results are also shown in many other scenarios, which will not be described here.

In short, we have verified the correctness of the simplified expression of ZZB in different common MIMO radar scenarios, which also illustrates the universality of this expression. When evaluating with simplified expression, the evaluation time of several days for an ordinary computer can be significantly shortened to few minutes without loss of accuracy, which provides a great help for judging the reliability of certain radar systems and the efficient preliminary system design.

## V. Conclusions

According to the results of the optimization problem obtained by simulation, we have simplified the original evaluation expression of ZZB. With the most time-consuming optimization part removed, a simplified expression was derived. It can be used to get results efficiently when judging the performance of a certain system. In the future work, we will extend this conclusion to the scenarios other than radar systems and further theoretically prove the correctness of this conclusion.

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